Gauss Elimination Method

The method is based on the idea of reducing the given system of equations **Ax = b,** to an upper triangular system of equations **Ux = z,** using elementary row operations. We know that these two systems are equivalent. That is, the solutions of both the systems are identical. This reduced system **Ux = z,** is then solved by the back substitution method to obtain the solution vector **x**.

We illustrate the method using the 3 × 3 system

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned} \tag{1}$$

We write the augmented matrix [**A | b**] and reduce it to the following form

$$[\mathbf{A}|\mathbf{b}] \xrightarrow{\text{Gauss elimination}} [\mathbf{U}|\mathbf{z}]$$

The augmented matrix of the system (1) is

$$\left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{array}\right] \tag{2}$$

First stage of elimination

We assume $a_{11} \neq 0$. This element a_{11} in the 1 × 1 position is called the *first pivot*. We use this pivot to reduce all the elements below this pivot in the first column as zeros. Multiply the first row in (2) by a_{21}/a_{11} and a_{31}/a_{11} respectively and subtract from the second and third rows. That is, we are performing the elementary row operations $R_2 - (a_{21}/a_{11})R_1$ and $R_3 - (a_{31}/a_{11})R_1$ respectively. We obtain the new augmented matrix as

$$\left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ 0 & a_{22}^{(1)} & a_{23}^{(1)} & b_2^{(1)} \\ 0 & a_{32}^{(1)} & a_{33}^{(1)} & b_3^{(1)} \end{array}\right] \tag{3}$$

where $a_{22}^{(1)} = a_{22} - \left(\frac{a_{21}}{a_{11}}\right)a_{12}$, $a_{23}^{(1)} = a_{23} - \left(\frac{a_{21}}{a_{11}}\right)a_{13}$, $b_2^{(1)} = b_2 - \left(\frac{a_{21}}{a_{11}}\right)b_1$,

$a_{32}^{(1)} = a_{32} - \left(\frac{a_{31}}{a_{11}}\right)a_{12}$, $a_{33}^{(1)} = a_{33} - \left(\frac{a_{31}}{a_{11}}\right)a_{13}$, $b_3^{(1)} = b_3 - \left(\frac{a_{31}}{a_{11}}\right)b_1$.

Second stage of elimination

We assume $a_{22}^{(1)} \neq 0$. This element $a_{22}^{(1)}$ in the 2 × 2 position is called the *second pivot*. We use this pivot to reduce the element below this pivot in the second column as zero. Multi-

ply the second row in (3) by $a_{32}^{(1)}/a_{22}^{(1)}$ and subtract from the third row. That is, we are performing the elementary row operation $R_3 - (a_{32}^{(1)}/a_{22}^{(1)})R_2$. We obtain the new augmented matrix as

$$\left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ 0 & a_{22}^{(1)} & a_{23}^{(1)} & b_2^{(1)} \\ 0 & 0 & a_{33}^{(2)} & b_3^{(2)} \end{array}\right] \quad (4)$$

where $a_{33}^{(2)} = a_{33}^{(1)} - \left(\frac{a_{32}^{(1)}}{a_{22}^{(1)}}\right) a_{23}^{(1)}$, $b_3^{(2)} = b_3^{(1)} - \left(\frac{a_{32}^{(1)}}{a_{22}^{(1)}}\right) b_2^{(1)}$.

The element $a_{33}^{(2)} \neq 0$ is called the *third pivot*. This system is in the required upper triangular form $[\mathbf{U}|\mathbf{z}]$. The solution vector $\mathbf{x}$ is now obtained by back substitution.

From the third row, we get $x_3 = b_3^{(2)}/a_{33}^{(2)}$.

From the second row, we get $x_2 = (b_2^{(1)} - a_{23}^{(1)} x_3)/a_{22}^{(1)}$.

From the first row, we get $x_1 = (b_1 - a_{12}x_2 - a_{13} x_3)/a_{11}$.

Procedure

if n is large it is difficult to find the solution by hand.

Augmented matrix $[A:B]$

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & b_n \end{array}\right]$$

Read $A(i,j), i = 1,\ldots,n\ j = 1,\ldots,n+1$

We will eliminate the first element in the *i*th row (except the first row)

$$a_{21} = 0 \Rightarrow \left(Row2 - \frac{a_{21}}{a_{11}} Row1 \right) \Rightarrow a_{21} - \frac{a_{21}}{a_{11}} a_{11} = 0$$

$$Row3 - \frac{a_{31}}{a_{11}} Row1$$

In general, let a_{i1} be the element of row i

$$Rowi - \frac{a_{i1}}{a_{11}} Row1, i = 2,\ldots,n$$

Ex.1: Solve the following linear system using Gaussian elimination:

$$x + 2y = 3$$

$$2x + 3y = 1$$

Transcribing the system of linear equations into an augmented matrix we obtain the matrix equation:

$$\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 2 & 3 & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 2 & 3 & 1 \end{array}\right] \xrightarrow{2R_1 \to R_1} \left[\begin{array}{cc|c} 2 & 4 & 6 \\ 2 & 3 & 1 \end{array}\right] \xrightarrow{R_2 - R_1 \to R_1} \left[\begin{array}{cc|c} 2 & 4 & 6 \\ 0 & -1 & -5 \end{array}\right]$$

We can clearly see from the resulting matrix that $-y = -5$ meaning that $y = 5$. We substitute this value into the equation equivalent to the first row of the resulting augmented matrix to solve for the variable x:

$$2x + 4y = 6 \to 2x + 4\,(5) = 2x + 20 = 6 \to 2x = -14 \to x = -7$$

Therefore, the final solution for the system of linear equations is:

$$x = -7 \quad and \quad y = 5$$

Ex.2: If we were to have the following system of linear equations containing three equations for three unknowns:

$$x + y + z = 3$$
$$x + 2y + 3z = 0$$
$$x + 3y + 2z = 3$$

We know from our lesson on representing a linear system as a matrix that we can represent such system as an augmented matrix like the one below:

$$\begin{array}{r} x + y + z = 3 \\ x + 2y + 3z = 0 \\ x + 3y + 2z = 3 \end{array} \quad \to \quad \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 1 & 2 & 3 & 0 \\ 1 & 3 & 2 & 3 \end{array}\right]$$

Let us row-reduce (use Gaussian elimination) so we can simplify the matrix:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 1 & 2 & 3 & 0 \\ 1 & 3 & 2 & 3 \end{array}\right] \xrightarrow{R_2 - R_1 \to R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & 2 & -3 \\ 1 & 3 & 2 & 3 \end{array}\right] \xrightarrow{R_3 - R_1 \to R_3}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & 2 & -3 \\ 0 & 2 & 1 & 0 \end{array}\right] \xrightarrow{2R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 2 & 4 & -6 \\ 0 & 2 & 1 & 0 \end{array}\right] \xrightarrow{R_3 - R_2 \to R_3}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 2 & 4 & -6 \\ 0 & 0 & -3 & 6 \end{array}\right] \begin{array}{r} x + y + z = 3 \\ 2y + 4z = -6 \\ -3z = 6 \end{array}$$

$$having : z = -2$$

$$2y + 4z = -6 \to 2y + 4(-2) = 2y - 8 = -6$$

$$2y = 2 \to y = 1$$

Applying the values of y and z to the first equation

$$x + y + z = 3 \to x + (1) + (-2) = x - 1 = 3 \to x = 4$$

And the final solution for the system is:

$$x = 4, \;\; y = 1, \;\; z = -2$$

Ex.3: Solve the following linear system using Gaussian elimination.

$$x + 3y + 3z = 2$$

$$3x + 9y + 3z = 3$$

$$3x + 6y + 6z = 4$$

We transcribe the linear system as an augmented matrix and then we start the Gaussian elimination process:

$$\left[\begin{array}{ccc|c} 1 & 3 & 3 & 2 \\ 3 & 9 & 3 & 3 \\ 3 & 6 & 6 & 4 \end{array}\right] \xrightarrow[R_2 - R_3 \to R_2]{} \left[\begin{array}{ccc|c} 1 & 3 & 3 & 2 \\ 0 & 3 & -3 & -1 \\ 3 & 6 & 6 & 4 \end{array}\right] \xrightarrow[\frac{1}{3}R_3 \to R_3]{}$$

$$\left[\begin{array}{ccc|c} 1 & 3 & 3 & 2 \\ 0 & 3 & -3 & -1 \\ 1 & 2 & 2 & 4/3 \end{array}\right] \xrightarrow[R_1 - R_3 \to R_3]{} \left[\begin{array}{ccc|c} 1 & 3 & 3 & 2 \\ 0 & 3 & -3 & -1 \\ 0 & 1 & 1 & 2/3 \end{array}\right] \xrightarrow[3R_3 \to R_3]{}$$

$$\left[\begin{array}{ccc|c} 1 & 3 & 3 & 2 \\ 0 & 3 & -3 & -1 \\ 0 & 3 & 3 & 2 \end{array}\right] \xrightarrow[R_3 - R_2 \to R_3]{} \left[\begin{array}{ccc|c} 1 & 3 & 3 & 2 \\ 0 & 3 & -3 & -1 \\ 0 & 0 & 6 & 3 \end{array}\right]$$

From which we can see that the last row provides the equation: $6z = 3$ and therefore $z = 1/2$. We substitute this in the equations resulting from the second and first row (in that order) to calculate the values of the variables x and y:

$$Having \;\; z = \tfrac{1}{2}$$

$$3y - 3z = -1 \to 3y - 3\left(\tfrac{1}{2}\right) = 3y - \tfrac{3}{2} = -1$$

$$3y = \tfrac{1}{2} \to y = \tfrac{1}{6}$$

Applying the values of y and z to the first equation:

$$x + 3y + 3z = 2 \to x + 3\left(\tfrac{1}{6}\right) + 3\left(\tfrac{1}{2}\right) = x + \tfrac{3}{6} + \tfrac{3}{2} = x + \tfrac{1}{2} + \tfrac{3}{2}$$

$$x + 2 = 2 \to x = 0$$

Therefore, the final solution to the system of linear equations is:

$$x = 0, \;\; y = \tfrac{1}{6} \;\; and \;\; z = \tfrac{1}{2}$$